the history of a 4.5-billion-year-old planet. If you were somehow dropped down on the Earth randomly at any moment in the past (or future), the chance of arriving at this critical moment would be less than 1 in 10 million. Our leverage on the future is high just now.

It might be a familiar progression, transpiring on many worlds—a planet, newly formed, placidly revolves around its star; life slowly forms; a kaleidoscopic procession of creatures evolves; intelligence emerges which, at least up to a point, confers enormous survival value; and then technology is invented. It dawns on them that there are such things as laws of Nature, that these laws can be revealed by experiment, and that knowledge of these laws can be made both to save and to take lives, both on unprecedented scales. Science, they recognize, grants immense powers. In a flash, they create world-altering contrivances. Some planetary civilizations see their way through, place limits on what may and what must not be done, and safely pass through the time of perils. Others are not so lucky or so prudent, perish.

Since, in the long run, every planetary society will be endangered by impacts from space, every surviving civilization is obliged to become spacefaring—not because of exploratory or romantic zeal, but for the most practical reason imaginable: staying alive. And once you're out there in space for centuries and millennia, moving little worlds around and engineering planets, your species has been pried loose from its cradle. If they exist, many other civilizations will eventually venture far from home.*

A MEANS HAS BEEN OFFERED of estimating how precarious our circumstances are—remarkably, without in any way addressing the nature of the hazards. J. Richard Gott III is an astrophysicist at Princeton University. He asks us to adopt a generalized Copernican principle, something I've described elsewhere as the Principle of Mediocrity. Chances are that we do not live in a truly extraordinary time. Hardly anyone ever did. The probability is high that we're born, live out our days, and die somewhere in the broad middle range of the lifetime of our species (or civilization, or nation). Almost certainly,